

National Informatics Centre (NIC)

Open Technology Group (OTG)

Professional Technical Architecture and CI/CD Pipeline Documentation

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Executive Summary

This document provides a comprehensive enterprise-level overview of the NIC OTG CI/CD implementation architecture, deployment workflow, automation strategy, and security integration. The solution enables secure, scalable, and automated software delivery using GitLab CI/CD, Apache2, PostgreSQL, SonarQube, PHPUnit, PHPStan, Trivy, and Gitleaks.

Project Overview

The project focuses on establishing a secure and enterprise-ready CI/CD ecosystem for PHP-based applications hosted under Apache2 infrastructure. The architecture integrates automated testing, code quality validation, vulnerability scanning, secret detection, deployment automation, and centralized reporting.

Technology Stack

The solution uses GitLab for source code management and CI/CD orchestration, GitLab Runner for distributed execution, PHP for backend application development, Apache2 for web hosting, PostgreSQL for database management, SonarQube for static code analysis, PHPUnit for testing, PHPStan for advanced type analysis, Trivy for vulnerability scanning, and Gitleaks for secret detection.

System Workflow

The workflow begins when developers push source code to GitLab repositories. The CI/CD pipeline automatically triggers validation stages including unit testing, static analysis, dependency audit, security scanning, vulnerability checks, deployment validation, and production deployment. Each stage ensures reliability, security, and compliance.

Enterprise Architecture

The enterprise architecture consists of Developer Systems, GitLab Repository Server, GitLab Runner Infrastructure, SonarQube Analysis Server, PostgreSQL Database Layer, Apache2 Deployment Server, Security Scanning Components, and Monitoring Systems. The architecture supports secure communication, distributed execution, and scalable deployment operations.

CI/CD Pipeline Architecture

The CI/CD pipeline includes multiple automated stages such as Unit Testing, Static Code Analysis, Dependency Audit, Secret Scanning, Vulnerability Scanning, Code Quality Validation, Database Validation, and Deployment. Parallel execution and caching mechanisms improve overall pipeline performance and scalability.

Security Architecture

The security architecture integrates multiple layers of protection including code quality enforcement, secret detection, dependency validation, vulnerability scanning, secure credential management, SSL/TLS communication, and security compliance verification. Security controls are enforced throughout the CI/CD lifecycle.

Deployment Architecture

Deployment automation is managed through GitLab CI/CD pipelines using secure runner execution. Application artifacts are deployed to Apache2 infrastructure with environment validation, rollback capability, deployment verification, and centralized logging.

Monitoring and Logging

The monitoring strategy includes pipeline logs, deployment logs, vulnerability reports, unit test reports, security scan reports, and centralized audit tracking. Logs are retained for traceability, troubleshooting, and compliance purposes.

Performance Optimization

Performance optimization is achieved using dependency caching, parallel pipeline execution, artifact reuse, optimized build processes, and selective scan execution. These techniques reduce deployment time and improve CI/CD efficiency.

Conclusion

The implemented enterprise CI/CD architecture provides a secure, scalable, and automated software delivery ecosystem suitable for NIC OTG technical operations. The integration of security, testing, deployment automation, and monitoring ensures high reliability and enterprise-grade operational excellence.